1. What I Have Explored and Built

I have always been curious about how Artificial Intelligence and Machine Learning work. To build a strong foundation, I studied neural networks, activations such as ReLU and GELU, attention mechanisms, Transformers, embeddings, and semantic search. I also followed Andrej Karpathy's educational implementations to understand Transformer internals, including multi-head attention, feed-forward networks, context modeling, and softmax-based attention.

My primary focus has been Retrieval-Augmented Generation (RAG). I have built RAG systems and gained hands-on experience with query routing, embedding and Large Language Model selection, schema extraction, metadata based retrieval, source attribution, multi-tenant system design, and efficient ingestion pipelines. I also implemented the HyDE technique from the paper *Precise Zero-Shot Dense Retrieval without Relevance Labels*, which showed me how research can significantly improve retrieval quality. While building AI pipelines on cloud services, I learned the importance of asynchronous processing, retry mechanisms, and robust system design for reliability and scalability.

2. Knowledge Gaps I Want to Address Through MLSS

Although I have worked on several AI projects, most of my experience is centered around LLM applications and retrieval systems. Through MLSS, I want to strengthen my understanding of core Machine Learning topics, with a particular focus on deep neural networks and reinforcement learning. I am eager to develop a deeper understanding of how neural architectures are designed, trained, and optimized, as well as how agents learn to make decisions through interaction and feedback. I also look forward to expanding my knowledge of related areas such as unsupervised learning, sequential learning, and causal inference.

I am particularly interested in understanding how models are optimized, evaluated, and scaled. While I have used frameworks such as RAGAS to evaluate AI systems, I want to learn the theory behind evaluation metrics, learning algorithms, and model design choices. I believe MLSS will help me bridge the gap between building AI applications and developing a deeper understanding of the Machine Learning principles that power them.

3. Why I Am a Strong Fit for MLSS

I believe I am a strong fit for MLSS because I learn by combining theory with implementation. Whenever I encounter a new concept or research paper, I try to understand the underlying ideas and apply them in practice. Implementing Transformer components and integrating HyDE into my RAG system reinforced my belief that the best way to learn AI is through building, experimenting, and iterating.

I am particularly interested in using research to solve real-world problems. I enjoy understanding the intuition behind new ideas and translating them into practical solutions. My experience applying techniques such as HyDE and studying concepts like ReLU and softmax has shown me how research can improve system performance. This curiosity, hands-on mindset, and commitment to continuous learning align strongly with the goals of MLSS, and I am excited to learn from experts, strengthen my fundamentals, and grow as an AI/ML engineer.

I look forward to this opportunity to deepen my knowledge and build intelligent systems that solve meaningful real world problems.